

REDDEPPA NAKKA

Software Engineer – Python, Backend & AI/ML

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Education

Rajiv Gandhi University of Knowledge Technologies, RK Valley

2021 – 2025

B.Tech in Computer Science and Engineering

CGPA: 8.5/10

Technical Skills

Languages: Python, JavaScript, Java, SQL, C# basics

Backend: FastAPI, Node.js, Express.js, Flask, REST APIs, JWT Authentication

Databases: PostgreSQL, MySQL, SQL Server, SQLite, MongoDB; Schema Design, Indexing

AI/ML: TensorFlow, Keras, Scikit-learn, NumPy, Pandas, OpenCV; LLM Integration, LangChain, RAG, Embeddings, Prompt Engineering, Agentic AI

Frontend: React 18, TypeScript, Vite, Tailwind CSS

Tools & Practices: Git/GitHub, GitHub Actions (CI/CD), Docker, Postman, Linux, Agile, Code Reviews, Unit & Integration Testing

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Experience

Befach 4X Pvt Ltd

May 2026 – Aug 2026

AI & Software Development Engineer Intern

Hyderabad, India

- Owned the majority of end-to-end development, in a two-person team, of an **AI-powered task execution platform** that converts multilingual meeting transcripts into structured, assignable tasks — shipped to production at taskmanager.befach.com.
- Designed and built the backend in **Node.js/Express with SQL**: REST APIs, relational schema design, JWT authentication with **role-based access control**, audit logging, real-time communication, and automated approval workflows.
- Engineered an **LLM task-extraction pipeline** (Claude API) producing schema-validated structured output — title, assignee, deadline, priority and ownership confidence — from **code-mixed English/Hindi/Telugu** transcripts, with a deterministic rule-based fallback that keeps the service available when the model API is unreachable.
- Extended the platform with **retrieval-augmented generation (RAG)**, conversational and voice-enabled AI, and natural-language deadline parsing; built the **React + TypeScript** client and customer-facing e-commerce flows with payment integration and search/filtering.

MAANG Technologies Pvt Ltd

May 2024 – Jan 2025

Software Engineering Intern

Remote

- Built RESTful backend services in **Python and FastAPI** serving **500+ requests/day**, covering endpoint design, data validation and error handling.
- Optimised service logic and validation paths, reducing average API response time by **40%** and feature delivery time by **30%**.
- Delivered **3 major feature releases** in 8 months within an agile team of **5+ engineers**, working through a GitHub pull-request and code-review workflow.

Projects

Atlas – Agentic AI Travel Recommender | [GitHub](#) | [Live](#) | Python, FastAPI, LangChain, Groq, React

- Built a FastAPI service running a **5-stage LangChain workflow** (intent extraction → filtering → cost estimation → seasonal ranking → recommendation) over **112 destinations across 62 countries**, with multi-currency parsing and graceful filter relaxation so every query returns grounded results.
- Cut cost and latency by **routing LLM stages by difficulty** — LLaMA 3.3 70B for reasoning, 3.1 8B for cheap parsing — with automatic failover to the fast model on rate limits, and integrated Amadeus live flight/hotel pricing behind a built-in estimator fallback.

Newsfall – Evidence-Linked Intelligence Pipeline | [GitHub](#) | Python, Next.js 15, Supabase Postgres + pgvector, GitHub Actions

- Automated a **daily 12-stage ingestion pipeline** (ingest → normalize → embed → entity/claim extraction → clustering → verification → scoring → briefing) on GitHub Actions cron, persisting to **Postgres with pgvector** for semantic search.
- Modelled a source-credibility registry and a **claim verification ladder** with per-source stance and contradiction detection, so every generated event links back to the evidence it was built from; served via Next.js 15 SSR.

Endometriosis Diagnosis – Classical ML & Deep Learning | [GitHub](#) | Python, Scikit-learn, TensorFlow, Keras, Flask

- Built a **soft-voting ensemble** (SVM + Gradient Boosting + Decision Tree) on clinical data reaching **92.7% accuracy**, 0.98 precision, 0.93 F1 on a 178-sample test set, cross-validated at **87.5%**; benchmarked it against hard voting, AdaBoost and each base classifier.
- Trained a **VGG16 + LSTM** hybrid for 4-class histopathology classification on 3,300 images (65% test accuracy, 0.88 precision on adenocarcinoma) and served it for real-time inference through a Flask app.

Achievements & Certifications

Artificial Intelligence & Machine Learning — certified course, IIT Patna

NPTEL Cloud Computing — **Elite + Silver**

Shipped an internship-built platform to production, now used by startups and small-business owners for team task tracking